Option Pricing Three Ways:

Black–Scholes Formula, Finite Differences and Trinomial Trees

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Abstract

A European call option is priced three ways: by the Black–Scholes formula, by solving the Black–Scholes partial differential equation (PDE) numerically, and on a trinomial tree. I show that the explicit finite-difference scheme and the trinomial tree are the same algorithm. Run on the same grid, they agree to 3×10^{-13}. The stability limit of the scheme is the condition that the tree's probabilities are non-negative. The explicit scheme converges at first order and Crank–Nicolson at second order. To reach a price error of 10^{-5}, Crank–Nicolson is about seven times faster. Crank–Nicolson produces a badly wrong gamma near the strike unless it is started with a few implicit (Rannacher) steps. Finally, I price American puts, where there is no formula. The early-exercise premium ranges from 1% to 25% of the European price across fifteen strikes and maturities, and I trace the optimal exercise boundary. All code, tests and figures are in the accompanying repository.

1 Introduction

The Black–Scholes model prices a European option with a formula. Most products traded in practice do not have one: American options, barriers and options under richer models all need numerical methods. The two standard approaches are *lattice methods* (binomial and trinomial trees), which are popular on trading desks because they are intuitive, and *finite-difference methods*, which solve the pricing PDE directly. This project builds both from scratch. I use the closed-form price as a benchmark and measure how accurate, how fast and how robust each method is.

The base case throughout is $S_0 = K = 100$, $T = 1$ year, $r = 5\%$ and $\sigma = 20\%$. For this case the call is worth 10.4506 and the put 5.5735.

2 The model and the PDE

Replication. The share follows geometric Brownian motion, $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$. Cash earns the risk-free rate r. Let $V(S,t)$ be the option's value. Form a portfolio that is long the option and short Δ shares. By Itô's formula,

$$\mathrm{d}(V - \Delta S) = \Big(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - \Delta\mu S\Big)\mathrm{d}t + \sigma S\,(V_S - \Delta)\,\mathrm{d}W_t.$$

Choosing $\Delta = V_S$ removes the random term. A riskless portfolio must earn the risk-free rate, so $\mathrm{d}(V - \Delta S) = r(V - \Delta S)\,\mathrm{d}t$. This gives the *Black–Scholes equation*

$$V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0, \qquad V(S,T) = \text{payoff}(S). \tag{1}$$

The drift μ has disappeared: hedging removes all dependence on the share's expected return. The companion project simulates exactly this.

Closed form. For a call, (1) has the solution $C = S\,N(d_1) - Ke^{-r(T-t)}N(d_2)$ with $d_{1,2} = \big[\ln(S/K)+(r\pm\frac{1}{2}\sigma^2)(T-t)\big]/\sigma\sqrt{T-t}$. The code also implements the Greeks. The tests check each one against bumped prices, and check that the Greeks satisfy (1) itself: $\Theta+\frac{1}{2}\sigma^2S^2\Gamma+rS\Delta-rV=0$ holds to 10^{-10}. In the base case $\Delta = 0.637$, $\Gamma = 0.0188$, vega $= 37.5$, $\Theta = -6.41$ per year and $\rho = 53.2$.

The log transform. The coefficients in (1) depend on S, which is awkward on a grid. With $x = \ln S$ and time to maturity $\tau = T - t$, the equation becomes

$$V_\tau = \tfrac{1}{2}\sigma^2 V_{xx} + \nu V_x - rV, \qquad \nu = r - \tfrac{1}{2}\sigma^2, \tag{2}$$

which has *constant* coefficients. It is a diffusion (heat) equation with drift and discounting, solved forward in τ from the payoff at $\tau = 0$.

3 Method 1: the explicit finite-difference scheme

Put a uniform grid $x_j = \ln S_0 + j\Delta x$, $\tau_n = n\Delta\tau$ and write $V_j^n \approx V(x_j, \tau_n)$. Use a forward difference in τ and central differences in x:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} = \frac{\sigma^2}{2}\,\frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{\Delta x^2} + \nu\,\frac{V_{j+1}^n - V_{j-1}^n}{2\Delta x} - rV_j^n.$$

Rearranging gives the explicit update

$$V_j^{n+1} = p_u V_{j+1}^n + p_m V_j^n + p_d V_{j-1}^n, \tag{3}$$

$$p_u = \frac{\lambda}{2} + \frac{\nu\Delta\tau}{2\Delta x}, \qquad p_m = 1 - \lambda - r\Delta\tau, \qquad p_d = \frac{\lambda}{2} - \frac{\nu\Delta\tau}{2\Delta x}, \qquad \lambda = \frac{\sigma^2\Delta\tau}{\Delta x^2}.$$

I fix λ and the number of time steps N. Then $\Delta\tau = T/N$ and $\Delta x = \sigma\sqrt{\Delta\tau/\lambda}$. The spot S_0 is placed exactly on a node, so no interpolation is needed. The grid extends to $\ln S_0 \pm 5\sigma\sqrt{T}$. Dirichlet boundary conditions are $V \approx 0$ (call) or $Ke^{-r\tau} - S$ (put) far below, and $S - Ke^{-r\tau}$ (call) or 0 (put) far above.

Accuracy. The local truncation error is $O(\Delta\tau + \Delta x^2)$. With λ fixed, $\Delta x^2 \propto \Delta\tau$, so the global error should be $O(\Delta\tau) = O(1/N)$.

Stability. If $p_u, p_m, p_d \geq 0$, each new value is a weighted average of old ones with weights summing to $1 - r\Delta\tau < 1$. Errors therefore cannot grow (a discrete maximum principle). The binding condition is $p_m \geq 0$, that is,

$$\lambda \leq 1 - r\Delta\tau \approx 1. \tag{4}$$

A von Neumann analysis gives the same answer. Substituting $V_j^n = g^n e^{ij\theta}$ gives an amplification factor $g(\theta) = p_m + (p_u + p_d)\cos\theta + i(p_u - p_d)\sin\theta$. The highest-frequency mode $\theta = \pi$ has $g = 1 - 2\lambda - r\Delta\tau$, and $|g| \leq 1$ again needs $\lambda \lesssim 1$. Once $\lambda > 1$, the sawtooth mode $(-1)^j$ is multiplied by roughly $2\lambda - 1 > 1$ at every step and blows up.

4 Method 2: the trinomial tree, which is the same thing

A recombining trinomial tree moves the log price up by Δx, leaves it unchanged, or moves it down by Δx over each step $\Delta\tau$, with risk-neutral probabilities q_u, q_m, q_d. Backward induction gives $V = D\,(q_u V_{\text{up}} + q_m V_{\text{mid}} + q_d V_{\text{down}})$ with discount factor D.

Proposition 1. *Let $q_k = p_k/(1 - r\Delta\tau)$ for $k \in \{u, m, d\}$ and $D = 1 - r\Delta\tau$. Then the tree reproduces the explicit scheme* (3) *exactly, and the q_k are probabilities if and only if the stability condition $p_k \geq 0$ holds. They match the first two moments of the log-price increment to leading order:*

$$(q_u - q_d)\Delta x = \frac{\nu\Delta\tau}{1 - r\Delta\tau}, \qquad (q_u + q_d)\Delta x^2 = \frac{\sigma^2\Delta\tau}{1 - r\Delta\tau}.$$

The proof is immediate: $p_u + p_m + p_d = 1 - r\Delta\tau$, and multiplying out gives (3). The point is the interpretation. The PDE scheme is a random walk under the risk-neutral measure, and *a negative middle probability is exactly what makes the scheme unstable.*

Why the agreement is exact. After N steps, the value at S_0 depends only on the $2N + 1$ nodes within N steps of it. This is the scheme's domain of dependence, and it is exactly the tree. If the grid has at least N nodes on each side of S_0, the boundary conditions never reach the answer. Table 1 confirms this for European calls and American puts: the largest difference across all runs is 2.8×10^{-13}, which is floating-point rounding. The more common tree discount $e^{-r\Delta\tau}$ differs from $1 - r\Delta\tau$ by $O(\Delta\tau^2)$ per step, which shifts the price by $O(\Delta\tau)$ in total. The convergence plot shows this as a curve sitting just below the explicit scheme's.

N	option	explicit FD	trinomial tree	|difference|
10	European call	10.376455406422	10.376455406422	4.6×10^{-14}
100	European call	10.443056135114	10.443056135114	8.5×10^{-14}
1000	European call	10.449829743431	10.449829743432	2.8×10^{-13}
10	American put	6.019863420594	6.019863420594	1.2×10^{-14}
100	American put	6.083551972862	6.083551972862	5.8×10^{-14}
1000	American put	6.089725322509	6.089725322509	1.1×10^{-13}

Table 1: Explicit scheme against trinomial tree, $\lambda = 0.5$, grid of half-width N.

5 Method 3: Crank–Nicolson

The θ-scheme averages the spatial operator $\mathcal{L}$ (the right-hand side of (2)) between the old and new time levels:

$$(I - \theta\Delta\tau\,\mathcal{L}_h)V^{n+1} = (I + (1 - \theta)\Delta\tau\,\mathcal{L}_h)V^n.$$

Here $\theta = 0$ is the explicit scheme, $\theta = 1$ fully implicit, and $\theta = \frac{1}{2}$ is *Crank–Nicolson* (CN). For $\theta > 0$ each step requires solving a tridiagonal linear system. This costs $O(J)$ operations for J grid points (`scipy.linalg.solve_banded`). In exchange, CN is stable for *any* $\Delta\tau$ and is second-order accurate in both $\Delta\tau$ and Δx.

Its weakness is the payoff kink at $S = K$. CN is not positivity-preserving: high-frequency modes are damped by a factor close to -1 per step when $\Delta\tau \gg \Delta x^2$. A kink contains every frequency, and the resulting oscillation is invisible in the price but ruins the second derivative, gamma. The standard fix is *Rannacher time-stepping*: replace the first CN step or two with fully implicit half-steps, which damp the high frequencies strongly, and then switch to CN.

6 Results

6.1 Convergence and cost

Figure 1 (left) plots the absolute pricing error against grid size. A least-squares fit on log–log axes gives slopes of -1.00 for the explicit scheme (against time steps) and -1.99 for Crank–Nicolson (against space steps), matching theory. For CN I used $N/4 + 1$ time steps for N space steps, because its stability does not tie $\Delta\tau$ to Δx^2.

The right panel shows the trade-off that matters in practice: error against CPU time. For a rough price, with an error of 10^{-3} (a tenth of a cent), the two are equally fast, at about 5 ms. For high accuracy, CN wins clearly. It reaches a 10^{-5} error with 3,200 space steps in about 0.1 s. The explicit scheme needs about 75,000 time steps for the same error and takes about 0.8 s, roughly seven times longer. The reason is the stability constraint. Halving Δx forces the explicit scheme to quarter $\Delta\tau$, so its cost grows like $N^{3/2}$.

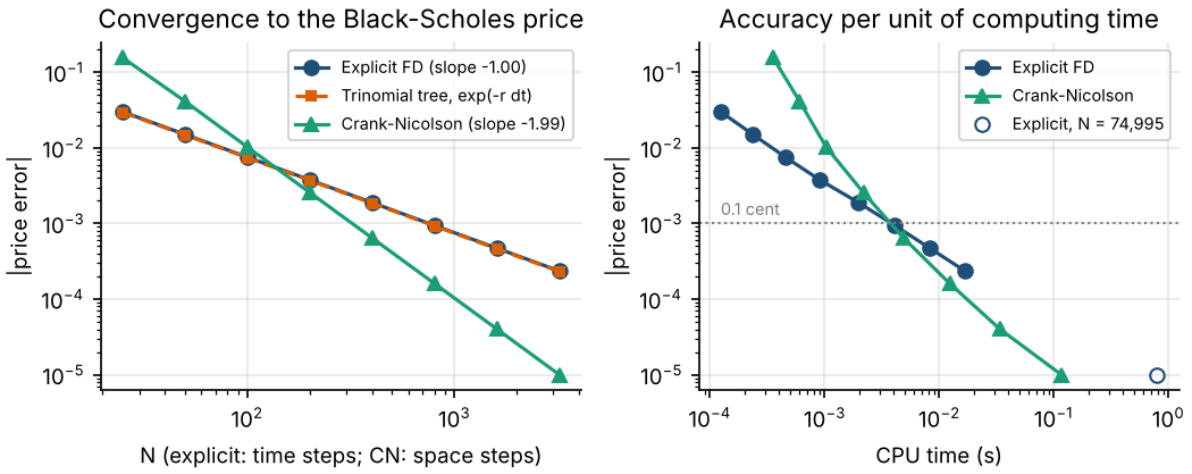


Figure 1: Left: error against resolution, with fitted slopes. Right: error against CPU time; the hollow point is the explicit scheme run until it matches CN's best accuracy.

6.2 Stability

Figure 2 runs the explicit scheme with 400 time steps for a range of λ. Below $\lambda = 1$ the error is small, between 1×10^{-4} and 2×10^{-3}. It is smallest near $\lambda \approx 0.9$, where the $O(\Delta\tau)$ and $O(\Delta x^2)$ error terms have opposite signs and partly cancel. At $\lambda = 1$, $p_m = -1.25 \times 10^{-4}$, only just negative, and the error is already 14 times larger than at $\lambda = 0.99$. At $\lambda = 1.01$ the error is 14, at $\lambda = 1.02$ it is 3×10^4, and at $\lambda = 1.05$ it is 10^{14}. The left panel shows the cause: the sawtooth mode grows outward from the strike, where the payoff kink seeded it. This is the mechanism the stability analysis predicts.

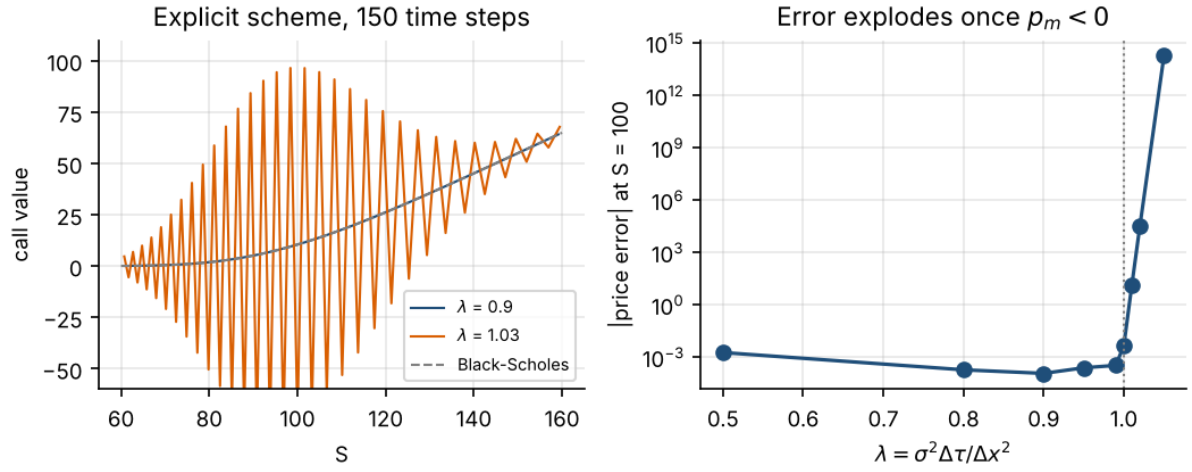


Figure 2: Left: the solution at $t = 0$ with $\lambda = 0.9$ (indistinguishable from Black–Scholes) and $\lambda = 1.03$ (150 steps). Right: price error against λ, 400 steps.

6.3 Greeks, and why Crank–Nicolson needs Rannacher

Greeks can be read off the grid at no extra cost. In $x = \ln S$, $\Delta = V_x/S$ and $\Gamma = (V_{xx} - V_x)/S^2$. With 3,200 explicit steps the grid gives $\Delta = 0.63684$ and $\Gamma = 0.018761$, against exact values of 0.63683 and 0.018762.

Figure 3 uses a harsher setting: a 3-month option with only 20 CN time steps on an 800-point grid. That is efficient for the price, but plain CN produces a gamma that swings between -0.6 and 1.2 around the strike, against a true value of about 0.04. The maximum gamma error near the strike is 1.15. Two Rannacher start-up steps reduce it to 2×10^{-5} at almost no extra cost. A risk system that reports gamma from a CN pricer without this fix would show large, spurious gamma exposure around every strike.

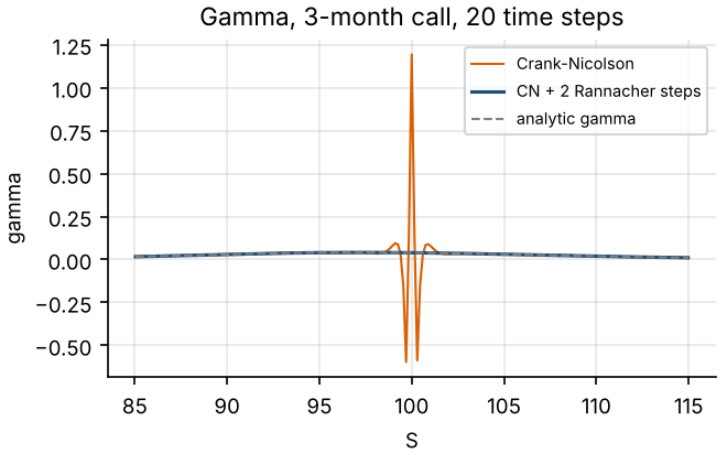


Figure 3: Gamma of a 3-month call from Crank–Nicolson with and without Rannacher start-up.

6.4 American puts

An American option can be exercised at any time, so its value satisfies $V \geq$ payoff everywhere. Holding is optimal where (1) holds; elsewhere exercising immediately is optimal. The boundary between the two regions, $S^*(t)$, is not known in advance (a free-boundary problem), and there is no closed form. On a tree or explicit grid the fix is a single line: after each backward step, replace V by $\max(V, \text{payoff})$. This is dynamic programming. At each node the holder compares the value of waiting with the value of exercising now.

Validation. The at-the-money one-year American put is worth 6.0901 (tree, 2,000 steps), against the published benchmark of 6.0904. An American *call* on a non-dividend paying share is worth exactly its European price in the code, as theory says. This is because $C \geq S - Ke^{-r\tau} > S - K$, so exercising early is never optimal.

Table 2 shows the premium rising with both strike and maturity. Deeper in-the-money puts and longer lives make it more valuable to receive K early and earn interest on it. The 3-month put struck at 120 is worth exactly 20.000, its intrinsic value: it should be exercised immediately. Figure 4 shows the exercise boundary for $K = 100$. Today the put should be exercised if $S \leq 81.0$. The boundary rises towards the strike as expiry approaches. The staircase shape comes from the grid spacing Δx.

T	K	European	American	premium	premium (%)
0.25	80	0.028	0.028	0.0003	1.2
0.25	100	3.373	3.480	0.107	3.2
0.25	120	18.709	20.000	1.291	6.9
1	80	0.687	0.724	0.036	5.3
1	100	5.574	6.090	0.517	9.3
1	120	17.395	20.136	2.741	15.8
2	80	1.507	1.670	0.163	10.8
2	100	6.611	7.723	1.112	16.8
2	120	16.509	20.681	4.173	25.3

Table 2: Early-exercise premium of American puts ($S_0 = 100$, $r = 5\%$, $\sigma = 20\%$). The full grid of fifteen cases is in `results/american_put_premium.csv`.

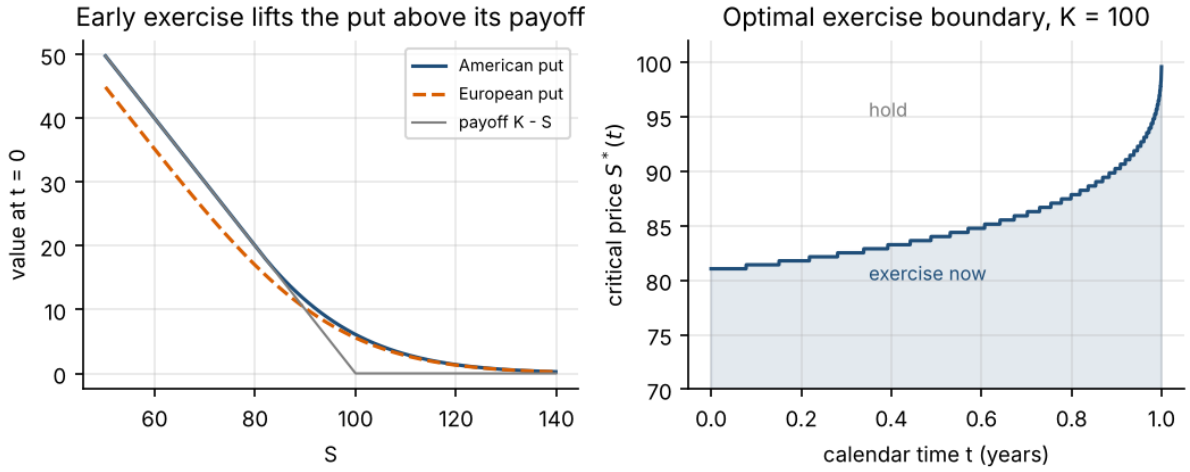


Figure 4: Left: American and European put values at $t = 0$. Right: the optimal exercise boundary $S^*(t)$, from the explicit scheme with 4,000 steps.

7 Discussion and limitations

- **Which method to use.** Trees are transparent and handle early exercise trivially, but they are explicit schemes and inherit first-order convergence and the stability constraint. Crank–Nicolson with Rannacher start-up is the better general-purpose tool. American options under CN need a projected solver (PSOR or a penalty method), which I have not implemented.
- **Strike placement.** The convergence study is at the money, where the payoff kink sits on a grid node. For other strikes the error oscillates as the grid is refined. The usual remedies are to align the grid with the strike or to smooth the payoff over one cell.
- **Model.** Volatility and rates are constant and there are no dividends. Discrete dividends would make early exercise of calls optimal in some cases, and local or stochastic volatility would make the PDE coefficients vary. The finite-difference code extends to both more naturally than the closed form does.

8 Conclusion

The three methods agree, and the reasons they agree are instructive. The explicit finite-difference scheme for the log-price Black–Scholes PDE is a trinomial tree under the risk-neutral measure. Its

stability condition is the requirement that the tree's probabilities be non-negative. Pushing past it destroys the solution within a few hundred steps. Crank–Nicolson removes the constraint and converges at second order, making it several times faster at high accuracy. It needs Rannacher smoothing to give usable Greeks. American puts carry an early-exercise premium of up to a quarter of the European price in the cases studied.

Reproducing the results. `pip install -e ".[test]"`, then `pytest` (20 tests) and `python scripts/run_experiments.py`. Timings are from a single CPU core and will vary by machine; the ratios should not.

References

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